(Please write your Enrollment Number) Enrollment No. 02701022022

**End-TermExamination**
**(CBCS)(SUBJECTIVE TYPE)(OffLine)**
**Course Name:B.Tech. (ECE, ECE-AI), Semester: 4th**
**(May, 2024)**

| Subject Code: BEC-202 | Subject: Linear Integrated Circuits (LIC) |
|---|---|
| Time : 3 Hours | Maximum Marks :60 |

**Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.**

| | | | | |
|---|---|---|---|---|
| **Q. 1** | | | **(2.5*8=20)** | |
| | (a) | Why an emitter follower has high input impedance? At what stage it is connected in internal circuit of op-amp? | | |
| | (b) | What is thermal drift in op-amp? What are different factors contributing to thermal draft? | | |
| | (c) | Give the output waveforms when the following inputs are given to a differentiator:- (i) Sine waveform (ii) Square waveform | | |
| | (d) | What are the conditions for Barkhausen criterion? | | |
| | (e) | Identify the type of circuit realizations using OTA given below in Fig 1(a) and Fig. 1(b). Give reasons in support of your answer. | | |

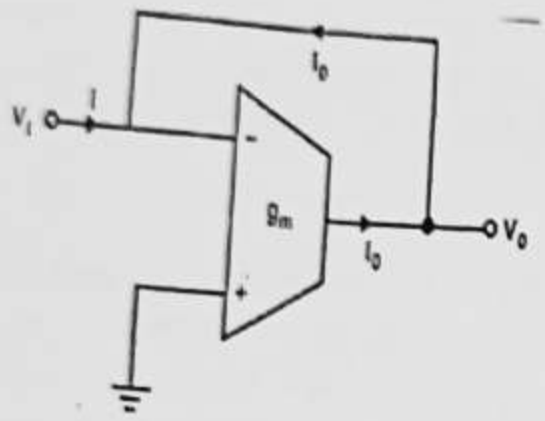



**Fig. 1(a)**

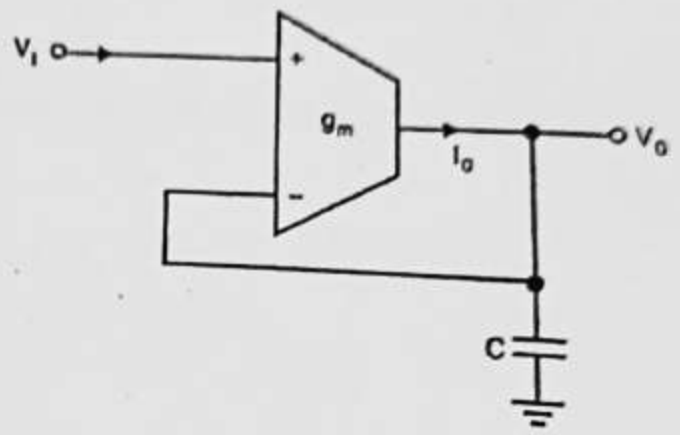



**Fig. 1(b)**

| | | | | |
|---|---|---|---|---|
| | (f) | What causes cross-over distortion? How it can be eliminated? | | |
| | (g) | What is damping coefficient? Give its significance in filter design. | | |
| | (h) | Define the following terms:<br>I. Lock-in range<br>II. Capture Range | | |
| | | | | |
| | | **UNIT-I** | | |
| **Q. 2** | Explain the following characteristic parameters of an op-amp with supporting diagrams:-<br>I. Input offset current<br>II. Input offset voltage<br>III. Total output offset voltage<br>IV. CMRR<br>V. Slew Rate | | **(10)** | |
| **Q. 3** | Give comparative analysis of different frequency compensation techniques for their frequency response curves along with their respective advantages and disadvantages. | | **(10)** | |
| | | **UNIT-II** | | |
| **Q. 4** | Explain the operation of triangular waveform generator with the help of circuit diagram and i/o waveforms. Also, derive the expression for its frequency of oscillation. | | **(10)** | |
| **Q. 5** | Explain the operation of logarithmic amplifier. Give detailed circuit diagram and mathematical analysis for the same. | | **(10)** | |
| | | **UNIT-III** | | |

| | | | |
|---|---|---|---|
| Q. 6 | Find out the transfer function of circuit given below in Fig. 2. | (10) | |

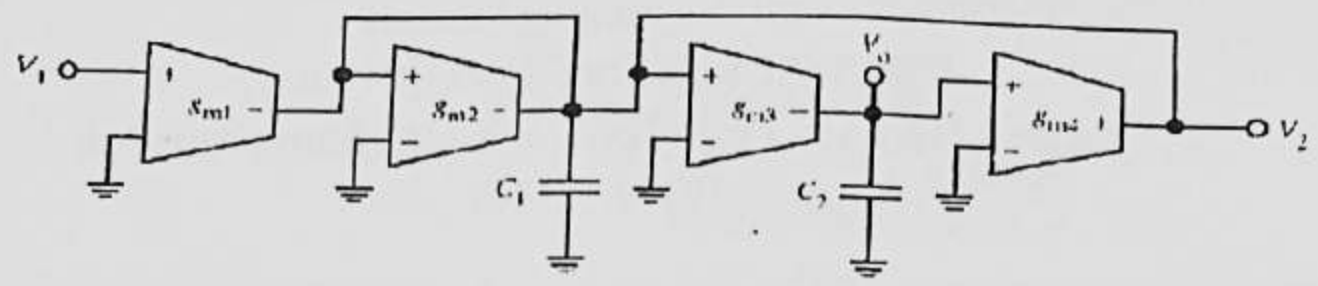



**Fig. 2**

| | | | |
|---|---|---|---|
| Q. 7 | What is the purpose of push-pull amplifier? Explain the operation of Class A push-pull amplifier with detailed circuit. List some advantages and limitations of a Class A push-pull amplifier? | (10) | |

**UNIT-IV**

| | | | |
|---|---|---|---|
| Q. 8 | Design a first order active wide band pass filter having $f_l$ = 400 Hz and $f_h$ = 2 kHz and pass band gain of 4. Give fully labeled circuit diagram with all components attached. Find the value of 'Q' of the filter. | (10) | |
| Q. 9 | Explain the operation of 555 Timer when it is connected for monostable operation. Draw functional diagram of monostable operation and its input, output and capacitor timing pulse waveforms. | (10) | |